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Thank you.

Biomedical Signal Processing and Control

A Comparative study of the latest Artificial intelligence-based Models for Plaque Measurement in a Carotid Ultrasound-based Japanese Cohort

--Manuscript Draft--

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Corresponding Author:	Jasjit S. Suri, Ph.D Atheropoint CA USA UNITED STATES
First Author:	Nikhil Singh
Order of Authors:	Nikhil Singh
	Rubeena Vohra
	Arun K. Dubey
	Pankaj K. Jain
	Mainak Biswas
	Ekta Tiwari
	Inder M. Singh
	Esma R Isenovic
	Amer M. Johri
	Laura E. Mantella
	John R Laird
	Narendra N. Khanna
	Zoltan Ruzsa
	Rajesh Singh
	Sanjay Saxena
	Mustafa Al-Maini
	Klaudija Viskovic
	J Fernandes E Fernandes
	Manudeep K Kalra
	Andrew Nicolaides
	George Kitas
	Luca Saba
	Vijay Viswanathan
	Jasjit S. Suri, Ph.D
Abstract:	Background and Motivation Accurate plaque measurement in carotid ultrasound images is essential for effective diagnosis and CVD risk stratification. Traditional methods for plaque assessment rely on manual segmentation and rule-based approaches, leading to inconsistent results and observer variability. Although deep learning (DL) models have increased segmentation accuracy, they face challenges in dealing with complex plaque structures

as well as low contrast ultrasound images. This study aims to compare state-of-the-art AI-driven models for carotid plaque measurement, evaluating their efficiency, robustness, and clinical applicability in a Japanese cohort.

Method

This study evaluates three deep learning models, i.e., (i) conventional neural networks(cNonAtt-dsCNN), (ii) attention-enhanced U-Net(cAtt-ssUNet),(iii) a two-stage attention-based method(cAtt-dsUNet) and compares against commercial AtheroEdge™ 2.0(AE2.0) from AtheroPoint, CA, USA-an automated method. All three DL models were trained on the same Japanese data with identical hyperparameters using the Adam optimizer, batch size 32, and a learning rate of 0.001 on an NVIDIA GPU cluster. The metrics used were: (i) total plaque area, and carotid intima-media thickness (cIMT), giving the automated lumen-intima, and media-adventitia (LIMA) boundaries in the far wall carotid ultrasound scans. The performance of these models was assessed using (i) Dice similarity, (ii) Jaccard index on binary segmented walls, and (iii) polyline distance metric for measuring cIMT.

Results

The results indicate the correlation between (i) plaque area to ground truth area cAtt-ssUNet<AE2.0<cAtt-dsUNet; (ii) for cIMT: cNonAtt-dsCNN<cAtt-dsUNet, (iii) for model size: AE2.0 <cNonAtt-dsCNN<cAtt-ssUNet<cAtt-dsUNet. (iv) Similarly, for dice similarity index cAtt-ssUNet<cAtt-dsUNet.

Conclusions

cAtt-dsUNet outperformed traditional DL methods and is comparable to the commercial system AtheroEdge™.

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Nikhil Singh^{1,2}, Rubeena Vohra¹, Arun K. Dubey¹, Pankaj K. Jain³, Mainak Biswas⁴, Ekta Tiwari⁵, Inder M. Singh², Esma R Isenovic⁶, Amer M. Johri⁷, Laura E. Mantella⁸, John R Laird⁹, Narendra N. Khanna¹⁰, Zoltan Ruzsa¹¹, Rajesh Singh¹², Sanjay Saxena¹³, Mustafa Al-Maini¹⁴, Klaudija Viskovic¹⁵, J Fernandes E Fernandes¹⁶, Manudeep K Kalra¹⁷, Andrew Nicolaides¹⁸, George Kitas¹⁹, Vijay Viswanathan²⁰, Luca Saba²¹, Jasjit S. Suri^{2,22,23,24}

¹Bharati Vidyapeeth's College of Engineering, New Delhi, INDIA.

²Stroke Monitoring and Diagnostic Division, AtheroPoint™, Roseville, California, USA.

³School of Medicine, Washington University, St Louis, MO USA.

⁴School of Computer Engineering, Kalinga Institute of Industrial Technology, Bhubaneswar, INDIA.

⁵Department of Innovation, Global Biomedical Technologies, Inc., Roseville, CA, USA.

⁶Department of Radiobiology and Molecular Genetics, "VINČA" Institute of Nuclear Sciences - National Institute of the Republic of Serbia, University of Belgrade, Belgrade, SERBIA.

⁷Division of Cardiology, Department of Medicine, Queen's University, Kingston, ON, CANADA.

⁸Department of Biomedical and Molecular Sciences, Queen's University, Kingston, ON, CANADA.

⁹Heart and Vascular Institute, Adventist Health St. Helena, St Helena, CA, 94574, USA.

¹⁰Department of Cardiology, Indraprastha APOLLO Hospitals, New Delhi 110001, INDIA.

¹¹Invasive Cardiology Division, University of Szeged, Szeged, HUNGARY.

¹²Department of Research and Innovation, Uttarakhand Institute of Technology, Uttarakhand University, Dehradun, 248007, INDIA.

¹³Department of CSE, IIIT, Bhubaneswar, INDIA.

¹⁴Allergy, Clinical Immunology and Rheumatology Institute, Toronto, ON M5G 1N8, CANADA.

¹⁵University Hospital for Infectious Diseases, Zagreb, CROATIA.

¹⁶Cardiovascular Institute and the Lisbon University Medical School, Hospital de Santa Maria, Lisbon 1600 190, PORTUGAL.

¹⁷Department of Radiology, Massachusetts General Hospital, Boston, MA, USA.

¹⁸Vascular Screening and Diagnostic Centre, University of Nicosia, Nicosia 2417, CYPRUS.

¹⁹Department of Rheumatology, University of Manchester, Dudley, UK.

²⁰MV Diabetes Centre, Royapuram, Chennai, Tamil Nadu, INDIA.

²¹Dept. of Neurology, University of Cagliari, Cagliari, ITALY.

²²Department of ECE, Idaho State University, Pocatello, ID, 83209, USA.

²³University Center for Research & Development, Chandigarh University, Mohali, INDIA.

²⁴Symbiosis Institute of Technology, Nagpur Campus, Symbiosis International (Deemed University), Pune, INDIA.

*Corresponding Author:

Dr. Jasjit S. Suri, PhD, MBA, FIEEE^a, FAIMBE^b, FSVM^c, FAIUM^d, FAPVS^e

^aFellow, The Institute of Electrical and Electronics Engineers

^bFellow, American Institute of Medical and Biological Engineering

^cFellow, Society of Vascular Medicine

^dFellow, American Institute of Ultrasound in Medicine

^eFellow, Asia Pacific Vascular Society

Stroke Monitoring and Diagnosis Division

AtheroPoint™, Roseville,

Roseville, CA 95661, USA

Phone: (916)-749-5628

Email: jasjit.suri@atheropoint.com